

Fundamentals of AI and KR - Module 3

2. Bayesian network representation

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Notice

Credits

The present slides are largely an adaptation of existing material, including:

- slides from [Russel & Norvig](#)
- slides by [Daphne Koller](#) on Probabilistic Graphical Models
- slides by Fabrizio Riguzzi on [Data Mining and Analytics](#)

I am especially grateful to these authors.

Downloading and sharing

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- Bayesian network representation

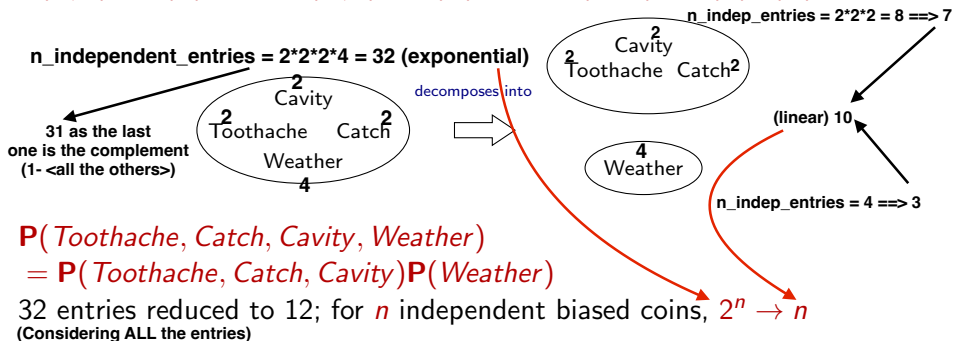
Independence



Independence

A and B are independent, denoted $\mathbf{P} \models (A \perp B)$, iff

$$\mathbf{P}(A|B) = \mathbf{P}(A) \quad \text{or} \quad \mathbf{P}(B|A) = \mathbf{P}(B) \quad \text{or} \quad \mathbf{P}(A, B) = \mathbf{P}(A)\mathbf{P}(B)$$



Absolute (marginal) independence powerful but rare. Independence is rare

Dentistry is a large field with hundreds of variables, none of which are independent. What to do?



Conditional independence

$P(\textit{Toothache}, \textit{Cavity}, \textit{Catch})$ has $2^3 - 1 = 7$ independent entries
 If I have a cavity, the probability that the probe catches in it doesn't depend on whether I have a toothache:

$$(1) P(\textit{catch}|\textit{toothache}, \textit{cavity}) = P(\textit{catch}|\textit{cavity})$$

The same independence holds if I haven't got a cavity:

$$(2) P(\textit{catch}|\textit{toothache}, \neg \textit{cavity}) = P(\textit{catch}|\neg \textit{cavity})$$

Catch is conditionally independent of *Toothache* given *Cavity*:

$$P(\textit{Catch}|\textit{Toothache}, \textit{Cavity}) = P(\textit{Catch}|\textit{Cavity})$$

Equivalent statements:

- $P(\textit{Toothache}|\textit{Catch}, \textit{Cavity}) = P(\textit{Toothache}|\textit{Cavity})$
- $P(\textit{Toothache}, \textit{Catch}|\textit{Cavity}) = P(\textit{Toothache}|\textit{Cavity})P(\textit{Catch}|\textit{Cavity})$

Notation: $P \models (\textit{Catch} \perp \textit{Toothache} | \textit{Cavity})$



Conditional independence

Write out full joint distribution using chain rule:

$$\begin{aligned} & \mathbf{P}(\textit{Toothache}, \textit{Catch}, \textit{Cavity}) \\ &= \mathbf{P}(\textit{Toothache} | \textit{Catch}, \textit{Cavity}) \mathbf{P}(\textit{Catch} | \textit{Cavity}) \mathbf{P}(\textit{Cavity}) \\ &= \mathbf{P}(\textit{Toothache} | \textit{Cavity}) \mathbf{P}(\textit{Catch} | \textit{Cavity}) \mathbf{P}(\textit{Cavity}) \end{aligned}$$

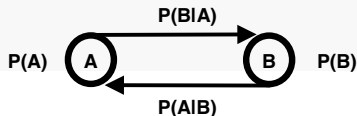
i.e., $2 + 2 + 1 = 5$ independent numbers

In most cases, the use of conditional independence reduces the size of the representation of the joint distribution from exponential in n to linear in n .

Conditional independence is our most basic and robust form of knowledge about uncertain environments.



Bayes' Rule



Product rule $P(a \wedge b) = P(a|b)P(b) = P(b|a)P(a)$

$$\Rightarrow \text{Bayes' rule } P(a|b) = \frac{P(b|a)P(a)}{P(b)}$$

or in distribution form

$$P(Y|X) = \frac{P(X|Y)P(Y)}{P(X)} = \alpha P(X|Y)P(Y) \quad \alpha = 1 / P(X)$$

Useful for assessing **diagnostic** probability from **causal** probability:

$$P(\text{Cause}|\text{Effect}) = \frac{P(\text{Effect}|\text{Cause})P(\text{Cause})}{P(\text{Effect})}$$



Example of diagnosis using Bayes' Rule

$$P(\text{Cause}|\text{Effect}) = \frac{P(\text{Effect}|\text{Cause})P(\text{Cause})}{P(\text{Effect})}$$

Say 1 individual in 50,000 suffers from meningitis, 1% from a stiff neck, and 70% of the times meningitis causes a stiff neck. *What is the probability that an individual with a stiff neck has meningitis?*

A = meningitis

B = stiff neck

P(A) = 1/50.000

P(B) = 0,01

P(B|A) = 0,70

P(A|B) = 0,70 * 0,00002 / 0,01 = 0,0014 = 0,14%



Example of diagnosis using Bayes' Rule

$$P(\text{Cause}|\text{Effect}) = \frac{P(\text{Effect}|\text{Cause})P(\text{Cause})}{P(\text{Effect})}$$

Say 1 individual in 50,000 suffers from meningitis, 1% from a stiff neck, and 70% of the times meningitis causes a stiff neck. *What is the probability that an individual with a stiff neck has meningitis?*

Let M be meningitis and S be stiff neck.

$P(m) = 1/50,000$, $P(s) = 0.01$, $P(s|m) = 0.7$.

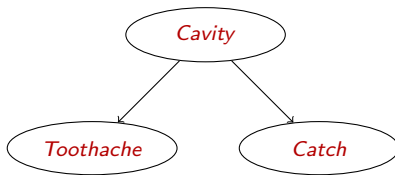
$$P(m|s) = \frac{P(s|m)P(m)}{P(s)} = \frac{0.7 \times 1/50,000}{0.01} = 0.0014$$

Note: posterior probability of meningitis still very small!



Bayes' Rule and conditional independence

$$\begin{aligned} & \mathbf{P}(Cavity | toothache \wedge catch) \\ &= \alpha \mathbf{P}(toothache \wedge catch | Cavity) \mathbf{P}(Cavity) \\ &= \alpha \mathbf{P}(toothache | Cavity) \mathbf{P}(catch | Cavity) \mathbf{P}(Cavity) \end{aligned}$$

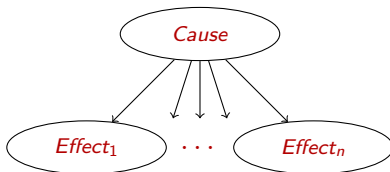




Bayes' Rule and conditional independence

conditional independence = independent given the cause

In the classification example we can consider the cause=class and each effect as a feature of the class, then we can compute the probability of a mix of features to be generated by a specific class (and then keeping the highest result) by computing $P(\text{class}) * \prod_i (P(\text{feature}_i | \text{class}))$



This is an example of a **naive Bayes** model:

$$P(\text{Cause}, \text{Effect}_1, \dots, \text{Effect}_n) = P(\text{Cause}) \prod_i P(\text{Effect}_i | \text{Cause})$$

Total number of parameters is linear in n



Summary so far

- Probability is a rigorous formalism for uncertain knowledge
- Joint probability distribution specifies probability of every atomic event
- Queries can be answered by summing over atomic events
- For nontrivial domains, we must find a way to reduce the joint size
- Independence and conditional independence provide the tools

Bayesian network representation



Bayesian networks

Probabilistic graphical model : DAG(Direct Acyclic Graphs) + UG(undirected graph)
like Bayesian Networks like Markov chains

A simple, graphical notation for conditional independence assertions and hence for compact specification of full joint distributions.

Syntax:

- a set of nodes, one per variable
- a directed, acyclic graph (link $\approx$ “directly influences”)
- a conditional distribution for each node given its parents:

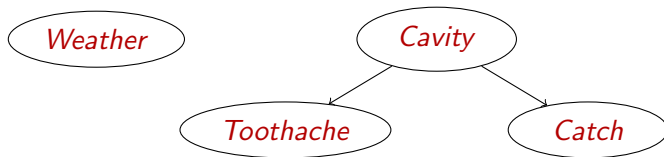
$$\mathbf{P}(X_i | \text{Parents}(X_i))$$

In the simplest case, conditional distribution represented as a **conditional probability table** (CPT) giving the distribution over X_i for each combination of parent values



Example

Topology of network encodes conditional independence assertions:



- *Weather* is independent of the other variables
- *Toothache* and *Catch* are conditionally independent given *Cavity*



Example

I'm at work, neighbor John calls to say my alarm is ringing, but neighbor Mary doesn't call. Sometimes it's set off by minor earthquakes. Is there a burglar?

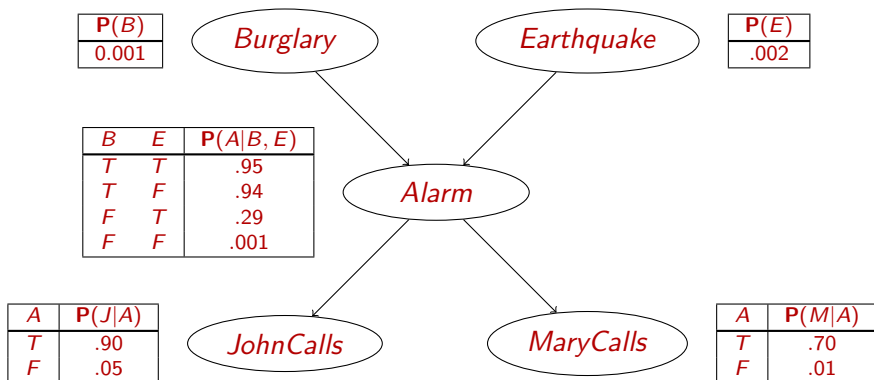
Variables: *Burglar*, *Earthquake*, *Alarm*, *JohnCalls*, *MaryCalls*

Network topology reflects "causal" knowledge:

- A burglar can set the alarm off
- An earthquake can set the alarm off
- The alarm can cause Mary to call
- The alarm can cause John to call



Example





Reasoning patterns

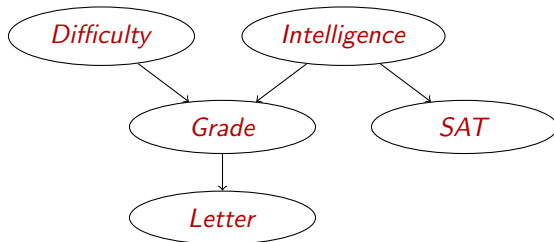
The student network

A student's **grade** depends on **intelligence** and on the **difficulty** of the course. **SAT** scores are correlated with intelligence. A professor writes recommendation **letters** by only looking at grades.

Reasoning patterns

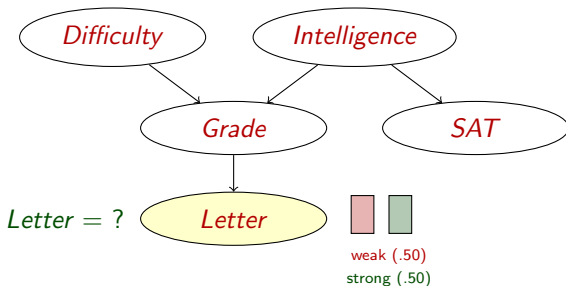
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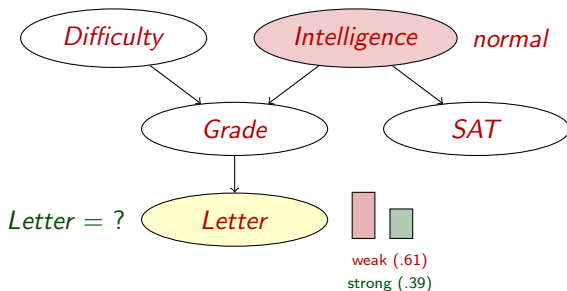
Reasoning patterns

- Causal: will George get a strong reference letter? (prediction)



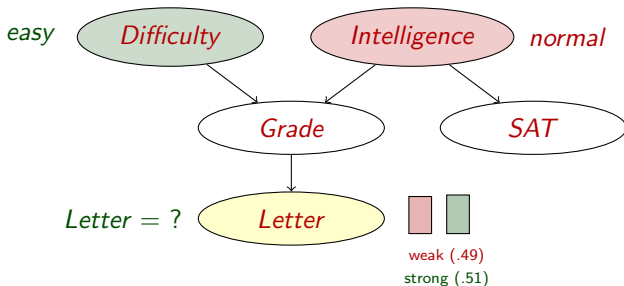
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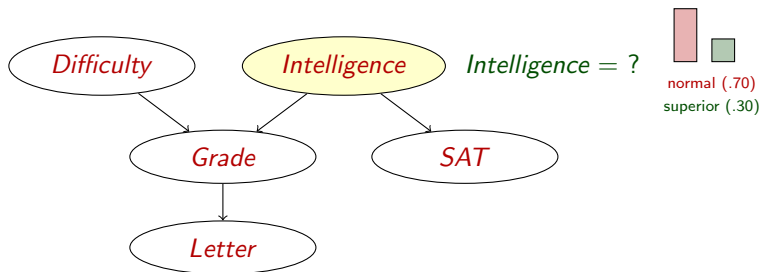
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Reasoning patterns

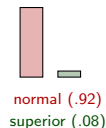
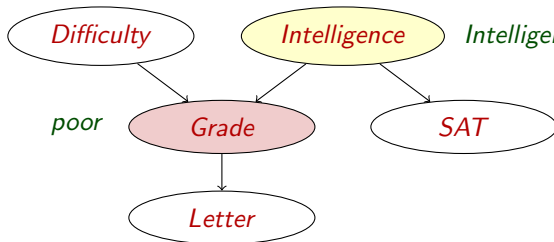
- **Causal**: will George get a strong reference letter? (**prediction**)
- **Evidential**: is George a good potential recruit? (**explanation**)



Reasoning patterns

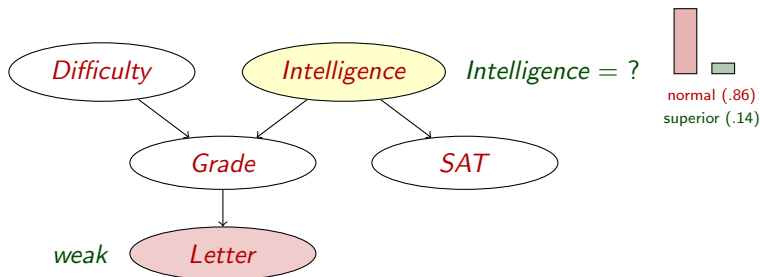
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Letter and intelligence are independent, since i know the grade



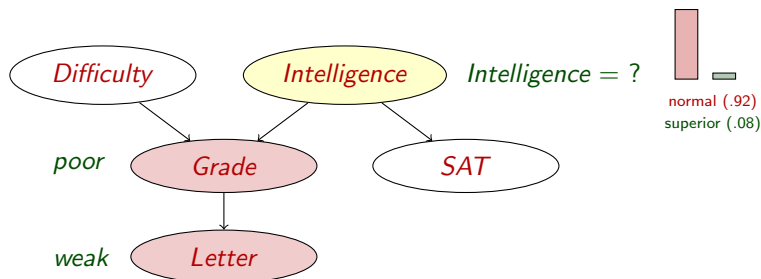
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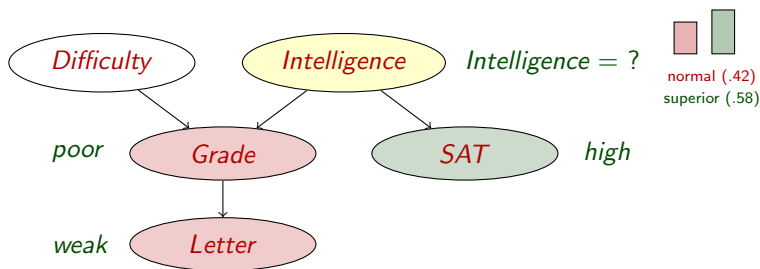
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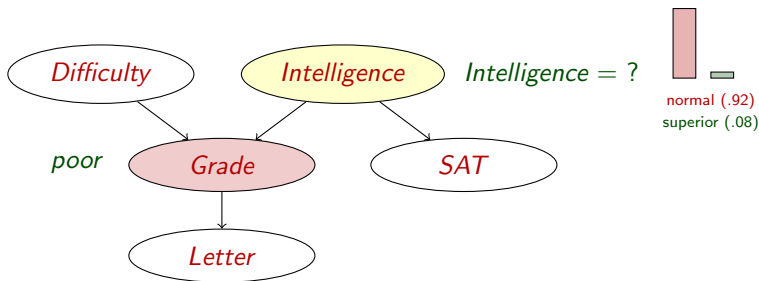
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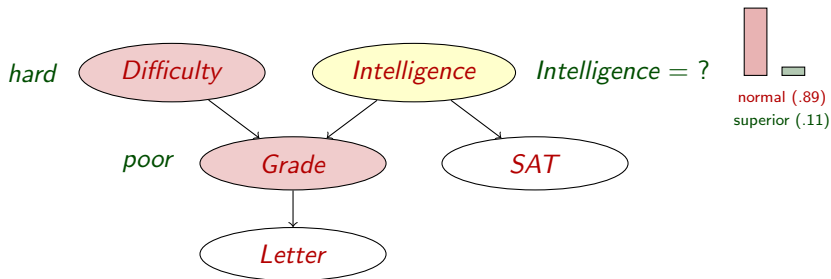
Reasoning patterns

- **Causal**: will George get a strong reference letter? (**prediction**)
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- **Intercausal**: why did George score low/high? (**explaining away**)



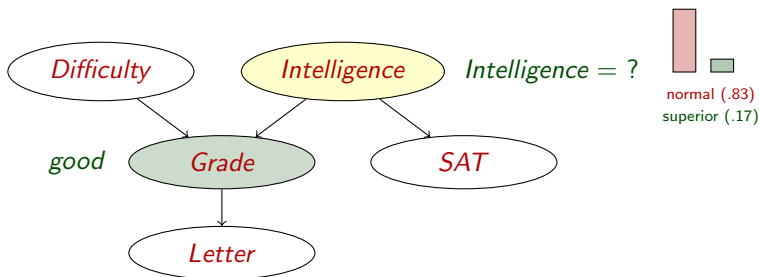
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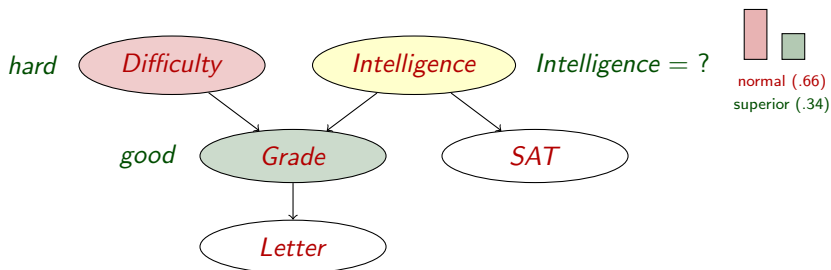
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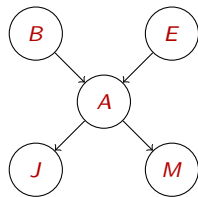




Compactness

A fully connected graph has the same power of a joint distribution.

A CPT for Boolean X_i with k Boolean parents has 2^k rows for the combinations of parent values. Each row requires one number p for $X_i = \text{true}$ (the number for $X_i = \text{false}$ is just $1 - p$)



If each variable has no more than k parents, the complete network requires $O(n \cdot 2^k)$ numbers

I.e., grows linearly with n , vs. $O(2^n)$ for the full joint distribution

For burglary net, $1 + 1 + 4 + 2 + 2 = 10$ numbers (vs. $2^5 - 1 = 31$)

For student net, $1 + 1 + 8 + 2 + 3 = 15$ numbers (vs. $2^4 \times 3 - 1 = 47$)

Full Joint Distribution (FJD)**Conditional Probability Table (CPT)**

Given $X_1, \dots, X_n$ variables
each with arity $d_1, \dots, d_n$

Then, given $d = \max\{d_i\}$,
the table will have complexity
 $O(d^n)$

Given $X_1, \dots, X_n$ variables
each with arity $d_1, \dots, d_n$
and with $k_1, \dots, k_n$ parents.

Then, given $d = \max\{d_i\}$
and $k = \max\{k_i\}$,
the table will have complexity
 $O(n \cdot (d-1) \cdot d^k)$

And since we assume $k \ll n$,
we are storing many less
numbers in this way

Global approach

Local approach



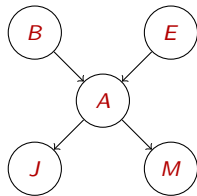
Global semantics

Global semantics defines the full joint distribution as the product of the local conditional distributions:

$$P(x_1, \dots, x_n) = \prod_{i=1}^n P(x_i | \text{parents}(X_i))$$

e.g., $P(j \wedge m \wedge a \wedge \neg b \wedge \neg e)$

$$= P(j \mid a) * P(m \mid a) * P(a \mid \neg b, \neg e) * P(\neg b) * P(\neg e)$$



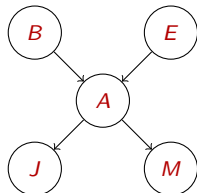


Global semantics

Global semantics defines the full joint distribution as the product of the local conditional distributions:

$$P(x_1, \dots, x_n) = \prod_{i=1}^n P(x_i | \text{parents}(X_i))$$

bottom-up exploration (child given the parent)

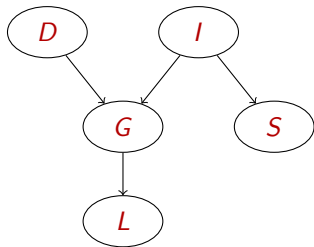


e.g., $P(j \wedge m \wedge a \wedge \neg b \wedge \neg e)$

$$\begin{aligned}
 &= P(j|a)P(m|a)P(a|\neg b, \neg e)P(\neg b)P(\neg e) \\
 &= 0.9 \times 0.7 \times 0.001 \times 0.999 \times 0.998 \\
 &\approx 0.00063
 \end{aligned}$$

Basic independencies in the Student network

S independent from **L** if we know **I**, because if we look for the influence of **S** on **L**, **I** is a far better indicator than **S** and so we don't need **S**.



What independencies?

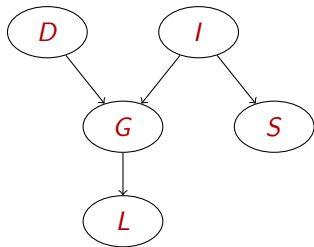
- $\mathbf{P} \models (L \perp \dots ?)$
- $\mathbf{P} \models (S \perp \dots ?)$
- $\mathbf{P} \models (G \perp \dots ?)$
- $\mathbf{P} \models (I \perp \dots ?)$
- $\mathbf{P} \models (D \perp \dots ?)$



L is not independent from **S**:

- **S** depends on **I**
- **L** depends on **G** which depends on **I**

Basic independencies in the Student network



What independencies?

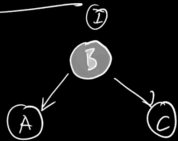
- $\mathbf{P} \models (L \perp I, D, S | G)$
- $\mathbf{P} \models (S \perp G, D, L | I)$
- $\mathbf{P} \models (G \perp S | I)$
- $\mathbf{P} \models (I \perp D)$ common effect ??
- $\mathbf{P} \models (D \perp I, S)$ common effect ??
- ...

Blue → starting node

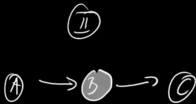
Yellow → node cutted off

Green → node that cuts off the yellow nodes

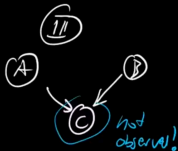
Basic Rules



$$P(A, C | B) = P(A | B) P(C | B)$$



$$P(A, C | B) = P(A | B) P(C | B)$$

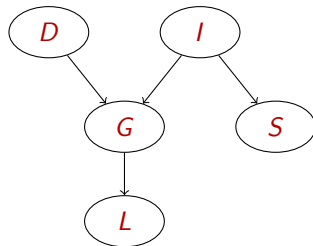


$$P(A, C) = P(A) P(C)$$

no given

"both A & B are roots"

Flow of probabilistic influence



Or else, when could X influence Y ?

- $X \rightarrow Y$ (direct cause)
- $X \leftarrow Y$ (direct effect)
- $X \rightarrow Z \rightarrow Y$ (causal trail)
- $X \leftarrow Z \leftarrow Y$ (evidential trail)
- $X \leftarrow Z \rightarrow Y$ (common cause)
- $X \rightarrow Z \leftarrow Y$ (common effect)

Definition (active two-edge trail)

If influence can flow from X to Y via Z , the trail $X \rightleftharpoons Z \rightleftharpoons Y$ is **active**



Flow of probabilistic influence: active trails

Consider a longer trail $X_1 \Rightarrow \dots \Rightarrow X_n$.

For influence to flow from X_1 to X_n , it needs to flow through every single node on the trail

This is true if and only if every two-edge trail $X_{i-1} \Rightarrow X_i \Rightarrow X_{i+1}$ along the trail allows influence to flow

Definition (active trail)

Let $\mathbf{Z}$ be a subset of observed variables.

The trail $X_{i-1} \Rightarrow X_i \Rightarrow X_{i+1}$ is **active** given $\mathbf{Z}$ if

- $\forall X_{i-1} \rightarrow X_i \leftarrow X_{i+1}$, X_i or one of its descendants are in $\mathbf{Z}$
- no other node along the trail is in $\mathbf{Z}$



Flow of probabilistic influence: direct separation

Definition (d-separation)

Two sets of nodes $\mathbf{X}$, $\mathbf{Y}$ are **d-separated** given $\mathbf{Z}$ if there is no active trail between any $X \in \mathbf{X}$ and $Y \in \mathbf{Y}$ given $\mathbf{Z}$

To determine if $\mathbf{X}$ and $\mathbf{Y}$ are **independent** given $\mathbf{Z}$:

- ① traverse the graph bottom-up marking all nodes in $\mathbf{Z}$ or having descendants in given $\mathbf{Z}$
- ② traverse the graph from $\mathbf{X}$ to $\mathbf{Y}$, stopping if we get to a **blocked** node
- ③ if we can't reach $\mathbf{Y}$, then $\mathbf{X}$ and $\mathbf{Y}$ are independent

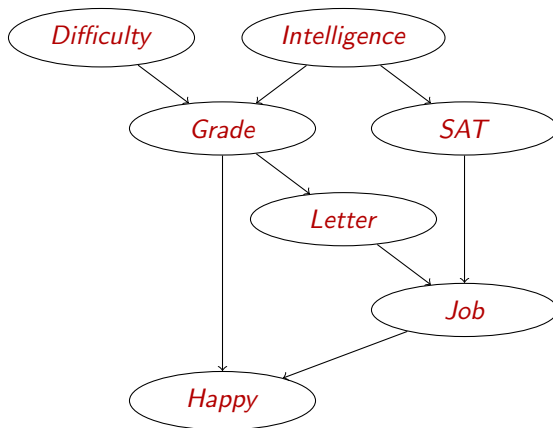
A node is **blocked** if either the middle of an unmarked v-structure, or in $\mathbf{Z}$ (not both)

in the
evidence

Example

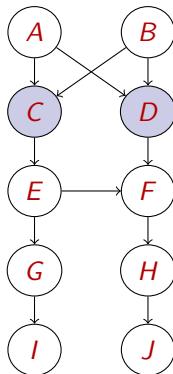
ancestral tree of a node := all the ancestor of a node

$(D \perp\!\!\!\perp J | L)$





Example

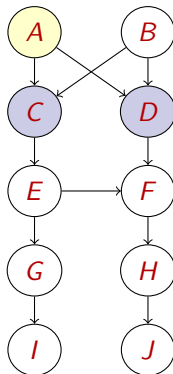


What independences?

- $\mathbf{P} \models (C \perp D)$?



Example

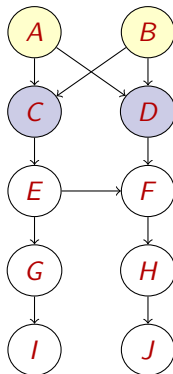


What independences?

- $\mathbf{P} \models (C \perp D)$?
- $\mathbf{P} \models (C \perp D | A)$?



Example

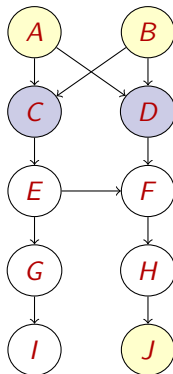


What independences?

- $\mathbf{P} \models (C \perp D)$?
- $\mathbf{P} \models (C \perp D | A)$?
- $\mathbf{P} \models (C \perp D | A, B)$?



Example

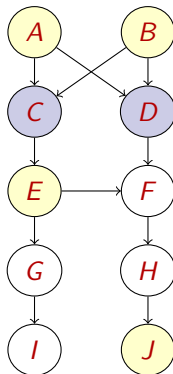


What independences?

- $\mathbf{P} \models (C \perp D)$?
- $\mathbf{P} \models (C \perp D | A)$?
- $\mathbf{P} \models (C \perp D | A, B)$?
- $\mathbf{P} \models (C \perp D | A, B, J)$?



Example



no $\rightarrow$ in V structure
C-E-F-D, F has a
descendent in the
evidence

What independences?

- $P \models (C \perp D)$? no
- $P \models (C \perp D | A)$? no
- $P \models (C \perp D | A, B)$? yes $\rightarrow$ V structure C-E-F-D
- $P \models (C \perp D | A, B, J)$?
- $P \models (C \perp D | A, B, E, J)$? Yes

Markov blanket

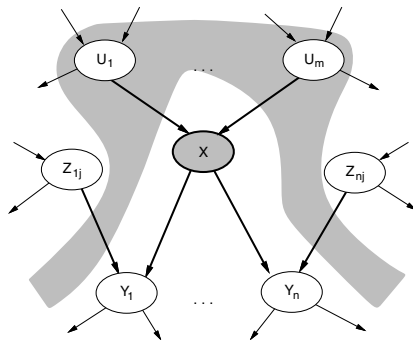
Given a node:

- If you have the parents you are blocking the upper flow
- If you have the children you are blocking the next flow



Local semantics

Local semantics: each node is conditionally independent of its nondescendants given its parents



If you know only the parents then you separates the node from everything except for its children

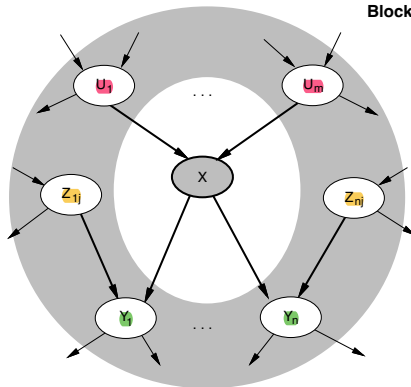
Theorem: **Local semantics** $\Leftrightarrow$ **global semantics**



Markov blanket

Each node is conditionally independent of all others given its **Markov blanket**: **parents** + **children** + **children's parents**

Blocking V-struct



If we know parents, children and children's parents we can infer everything about X ignoring the rest of the network

Questions?